RESULTS: PKM fall-degree study, the t = 13 bracket (jobs pfd-x-t13, Amendment 7 B)

Reading (applied by hand from the rule text; the Amendment 7 D scorer patch is not committed): at Kummer m = 2, P1, t = 13, c*(13) = 6. Draw 1 gives d_F = 6. Draw 0 gives only d_F >= 6. Amendment 7 D.5 reads **"undecided at t = 13"**. It is not "holds at t = 13". [pcc:unverified]

- Draw 1: cap 5 does not refute and cap 6 refutes (inconsistent true). So d_F = 6 = c* for draw 1.
- Draw 0: cap 5 does not refute, so d_F >= 6. Cap 6 has no row: the cell was launched three times and each launch was killed by an EC2 Spot interruption. Draw 0 has a one-sided bound only.
- The median over the two draws is a bound that includes c* = 6. D.5 calls that "undecided".
- One draw agrees with c*. The other is unresolved for lack of a completed run. No row measures a disagreement with c*.
- Both draws have grid_zeros = 0: neither is decomposable.

Every number below is [pcc:unverified]: no pcc claim id exists for any of them.

Evidence classes: **measured** = read from the rows under results/Y1/, results/Y2/ and results/Y4/; **derived** = a comparison or the floor in c* applied to those rows by the scratchpad script of the archive in section 6.

1. What was run (measured)

Three command jobs completed, one capped F4 run each (f4_fp_tower), on Iso Lab EKS Spot, on the frozen image and binaries of pfd-a. The cell is Kummer m = 2, p = 2013265921 (P1), t = 13 (N = 26), random targets, draws 0 and 1 (target_index), --budget 86400.

job	card (launch)	draw	cap	request file	--max-dense	memory	protocol_sha256 (request)
pfd-x-t13-k0-c5-20261108 (Y1)	Y1-1	0	5	requests_r2g/pfd-x-t13-k0-c5.json	27,900,000,000	112Gi	bbd17530ee06205a...
pfd-x-t13-k1-c5-20261108 (Y2)	Y2-1	1	5	requests_r2g/pfd-x-t13-k1-c5.json	27,900,000,000	112Gi	bbd17530ee06205a...
pfd-x-t13-k1-c6-20261108 (Y4)	Y4-1	1	6	requests_r2h/pfd-x-t13-k1-c6.json	12,800,000,000	56Gi	a10b9393ef23d5e7...
pfd-x-t13-k0-c6 (draw 0, cap 6)	Y3-1, Y5-1, Y6-1	0	6	see section 2a			

- **Two memory sizes, two protocol hashes.** The cap-5 jobs were submitted at 112Gi (Amendment 7 A). The cap-6 job of draw 1 was submitted after Amendment 14 and asks for 56Gi with --max-dense 12800000000. Each receipts/Y*/launch/request.json equals the request file named in the table (compared as JSON, derived). Amendment 14 says the limit only stops a run and does not steer it; the two cap-5 rows and the cap-6 row each end below their own limit, so the limit did not act in any of them (section 2b).
- **Resources per request:** cpu 12, seconds 90000, isolation_profile standard, retries 0, no instance pin. host.txt of every pod: nproc 12, rayon: 12.
- **Image and binaries:** 104186811093.dkr.ecr.us-east-2.amazonaws.com/isolab/tasks@sha256:0d6b2ecd6df3f5f092fe4b6179986b041eb7d6be3b94e52d92809bdab2ff3ae8. binary.sha256 (each pod): source revision f13a690df016d0a457bd40579fbb9d054f486adf, pkm_tower_pilot f78f3959..., prime_fall_degree_tools 7e8a6703..., pfd_grid_zeros 962833611.... These equal the t = 12 note's.
- **Node types** (host.txt and SPOT_LOSSES.md):

job	node type	CPU (host.txt)	MemTotal	pod start to end (UTC, progress.txt)	pod
Y1	r6i.4xlarge (SPOT_LOSSES.md)	Intel Xeon Platinum 8375C	129804652 kB	2026-11-10 03:09:48 to 03:10:05	9253a575...
Y2	not recorded in SPOT_LOSSES.md	AMD EPYC 7R13	129114476 kB	03:19:25 to 03:19:42	e7e4ad4c...
Y4	m8i.4xlarge (SPOT_LOSSES.md)	Intel Xeon 6975P-C	64776640 kB	07:12:18 to 10:25:08	2f855278...

For Y2, any instance type named from the CPU model is an inference. - **Cards and receipts.** Launch cards Y1-1, Y2-1 and Y4-1 and collect cards Y1-C01, Y2-C01 and Y4-V01 have receipts with status completed (cards/*.receipt.json). The jobs are recorded in receipts/Y1/, receipts/Y2/ and receipts/Y4/.

2. The three rows (measured)

Source: results/Y{1,2,4}/<pod>/runs/X-m2-t13-k<draw>-c<cap>.jsonl. grid_zeros from the .gridzeros.jsonl beside each.

row	draw	cap	degree_reached	inconsistent	basis_len	timed_out	oversize	max_cols	max_dense_entries	muladds	wall_s	grid_zeros	refutes
Y1	0	5	5	false	2922	false	false	16572	53833099	179052499158	16.8	0	no
Y2	1	5	5	false	2922	false	false	16572	53833099	179052499158	16.8	0	no
Y4	1	6	6	true	1	false	false	177063	4717115340	288586444518022	11570.0	0	yes
(none)	0	6	no row										no row

- Each row has zero_dim true, full_rank_exit true, planted_ok null (random target), and its x_r equals the one in the matching .gridzeros.jsonl row (1118191946 for draw 0, 1132329660 for draw 1).
- **Compared with t = 12 (derived).** The cap-5 rows Y1 and Y2 equal X1 and X2 of the t = 12 note in max_cols, max_dense_entries, muladds, basis_len and steps (22). Wall time differs (16.8 s here, 10.8 s and 11.8 s at t = 12, on other pods). The cap-6 row Y4 differs from X4: max_cols 177063 against 103785, max_dense_entries 4717115340 against 1357804954, steps 32 against 30, wall_s 11570.0 against 1762.8. Both end with basis_len 1 and inconsistent true.

2a. Draw 0, cap 6: no row (measured)

- The cell pfd-x-t13-k0-c6 was launched three times and killed each time by an EC2 Spot interruption: pfd-x-t13-k0-c6-20261108 (card Y3-1, r7a.4xlarge, about 10 min), pfd-x-t13-k0-c6-20261110-r1 (card Y5-1, r6a.4xlarge, about 20 min) and pfd-x-t13-k0-c6-20261110-r2 (card Y6-1, 56Gi under Amendment 14, m7i.4xlarge, 3 h 51 min). SPOT_LOSSES.md (rows 1 to 3 and the section on the cell) has the times, the audit-log events and the node types.

- Each attempt wrote no result row (the row files in S3 are 0 bytes). **Nothing from the three lost attempts is admitted or analysed in this note.** I did not read their partial logs for it.
- Owner decision 3 allowed one relaunch; owner decision 7 allowed one third attempt. Both were used. The cell is back with the owner.

2b. Amendment 14 conditions for the 56Gi run Y4 (measured, then compared)

condition (Amendment 14, sections 3 and 4)	Y4	holds
memory request 56Gi, `--max-dense 12800000000`	`argv: cap:13:1:6:12800000000`; request `memory "56Gi"`; `MemTotal` 64776640 kB	yes
`oversize false`	`oversize false`	yes
`timed_out false`	`timed_out false`	yes
`max_dense_entries` below the limit	4,717,115,340 against 12,800,000,000 (about 37 percent of the limit)	yes
not killed by the memory limit (`OOMKilled`)	exit 0, job state `succeeded`; the run wrote its row	yes
peak memory	20,618 MB (the peak value in the F4 log lines, largest over the log)	recorded

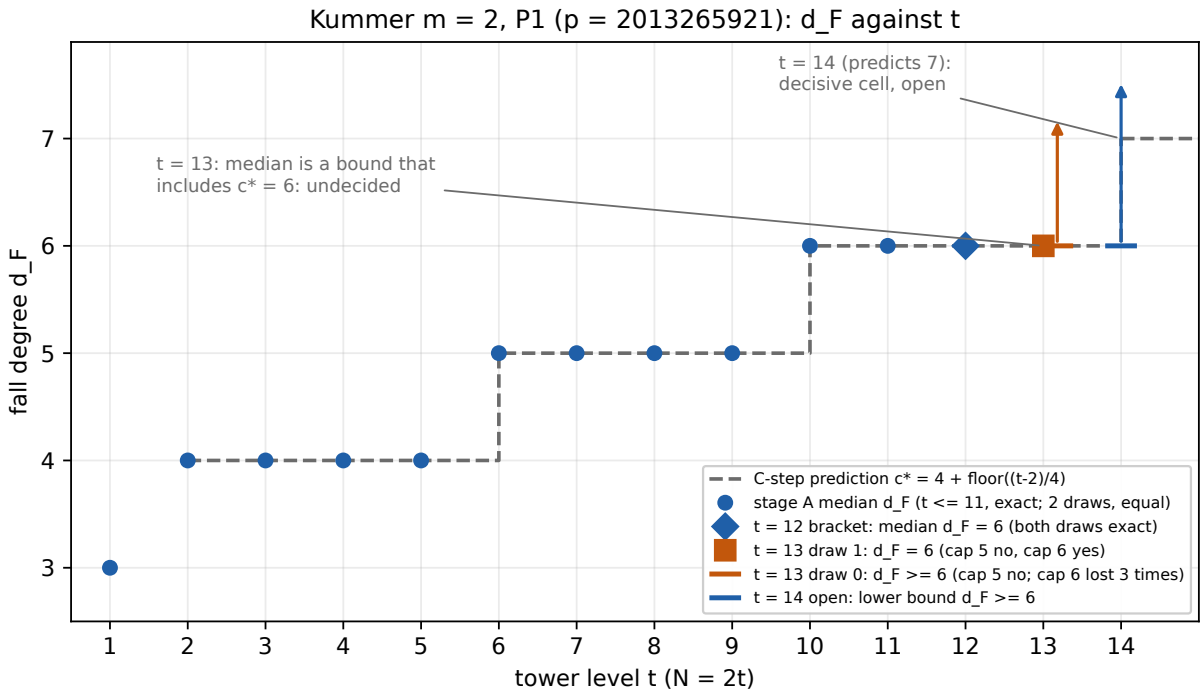
- By Amendment 14 section 4, a 56Gi run that completes with `oversize false` and `timed_out false` is the cell's row, whatever it reads. No `-m112` re-run applies to Y4.
- Peak memory of 20,618 MB is about 32 percent of the pod's `MemTotal` (derived).

3. The bracket and the D.5 line (derived, by hand)

Amendment 7 B at `t = 13`: `c* = 4 + floor((13 - 2)/4) = 6`. Run cap `c* - 1 = 5`; if it does not refute, run cap `c* = 6`; if that refutes, `d_F = c* = 6`.

draw	cap 5	cap 6	d_F
0	does not refute (Y1)	no row (lost three times)	>= 6
1	does not refute (Y2)	refutes (Y4)	6

- Draw 1 is exact and uncensored: no `timed_out`, no `oversize`.
- Draw 0 is a lower bound only. The rule has no value for a draw whose cap-6 run did not finish for a reason outside the computation.
- The median of a draw with `d_F >= 6` and a draw with `d_F = 6` is a bound that includes `c* = 6`. Amendment 7 D.5 ("undecided at t": a bound that includes c*, or no median). **Reading: undecided at t = 13.**
- It is not "holds at t = 13". "Holds" needs a median that equals c*, and the median here is not exact.
- Plainly: draw 1 agrees with c*. Draw 0 is unresolved because its run never completed. It is not unresolved because of any measured disagreement.
- Stage A's row for `t = 13` (closure one-sided, `>= 6`) is unchanged in its reading. The `t = 12` row is as in the t = 12 note ("holds at t = 12"). `t = 14` is unchanged.



Sources: stage A pfd-a-20261108 (t <= 11), X1-X4 (t = 12), Y1, Y2, Y4 (t = 13), results/Y13/analysis/pfdY13_summary.json. n = 2 draws per t, none decomposable. [pcc:unverified]

Figure 1: d_F against t, Kummer m = 2, P1. Sources: pfd-a-20261108 (t <= 11), jobs X1-X4 (t = 12), Y1, Y2, Y4 (t = 13); n = 2 draws per t. [pcc:unverified]

4. What this shows and does not show

Shows (all `[pcc:unverified]`): on draw 1 at `t = 13`, capped F4 at degree 5 does not reach 1 and capped F4 at degree 6 does. On draw 0, capped F4 at degree 5 does not reach 1. Under the Amendment 7 B and D.5 rule text, the cell reads "undecided at t = 13". The one exact draw equals the registered C-step value 6.

Does not show, each stated once:

- **One draw only.** Exactly one draw has a completed bracket. A single value has no spread, and no median over two draws exists.

- **The lost cell.** Draw 0 at cap 6 has no row. Whether cap 6 refutes on draw 0 is not known from any admitted data. The owner decides whether the lost cell gets another attempt. Owner decision 7 allowed one third attempt, and that attempt was lost.
- **Not a test that separates the readings.** `c*(13) = 6`, the value already seen at `t = 11` and `t = 12`. This cell cannot tell "C-step" from "flat at 6". `t = 14` is the level where `c*` changes to 7. It is the decisive cell (Amendment 7 E) and has no row.
- **Scorer.** The D.5 reading was applied from the rule text by hand and by a scratchpad script. The Amendment 7 D scorer patch is still not committed or reviewed (RESULTS.md section 8). The committed `prime_fall_degree_tools score` was not run on these rows.
- **G3.** G3 (capped F4 agrees with the closure where both run) passed in `pfd-a` (RESULTS.md section 1) and in the stage C G3-only job `pfd-g3c-20261109` (RESULTS_STAGE_C.md section 2a), on cells with `m <= 6`. G3's tool tests `N <= 20`; `N = 26` here is outside that range, so no G3 run covers this cell itself.
- **Scope of the measurement.** This is a fall-degree measurement on toy primes (`P1 = 2013265921`), `m = 2` only. It is not an index-calculus cost result: no leaderboard row, no rho comparison, no `S`, and no gain over the index-calculus baseline is claimed. `[pcc:unverified]`

5. Limits

- Wall seconds depend on the host and are context only; no timing claim is made. The three rows ran on three different node types. No L2, replay or A/A admission applies to these command jobs (README).
- The node type of Y2 is not recorded.
- The cap-5 rows ran at 112Gi and the cap-6 row at 56Gi (Amendment 14). The cap-5 rows equal their `t = 12` counterparts in every compared count, and Y4 ended well below its limit, so the size does not show in the data.
- Peak memory is read from the F4 log lines, not from a pod metric.
- The draw 0 cap-6 cell has no row (section 2a).
- No pcc claim was filed. Every number is an unverified assertion until a claim and a signed verdict exist.

6. Evidence paths

Under `research/pkm_falldegree_unblock_20261107/`:

- Rows: `results/Y1/9253a575-a872-49e5-886e-eaa87e0ff902/`, `results/Y2/e7e4ad4c-2d4b-4f65-96b9-6b10ededf985/`, `results/Y4/2f855278-d366-4b77-a295-e2a784e950ba/` (each: `runs/`, `host.txt`, `progress.txt`, `binary.sha256`). These directories are git-excluded; the raw files are in S3 by SHA-256 through the benchmark record below.
- Requests: `requests_r2g/pfd-x-t13-k{0,1}-c5.json`, `requests_r2h/pfd-x-t13-k1-c6.json`.
- Receipts: `receipts/Y1/`, `receipts/Y2/`, `receipts/Y4/`; `cards/CARD-pkm-falldegree-unblock-20261107-pfd-x-t13-*.receipt.json`.
- Spot losses: `SPOT_LOSSES.md`.
- `t = 12` rows compared: `results/X1..X4/`; `results/X12/analysis/pfdX12_summary.json`. Stage A values for `t <= 11`: `results/R9/analysis/pfdA_summary.json`.
- Benchmark record: `benchmarks/records/20261110T0305Z-pkm-fd-pfd-x-t13.json` (commands and raw entries; the scripts tarball was added to it, nothing else).
- This analysis: `results/Y13/analysis/` holds `pfdY13_fig1.svg` and `.png` (`d_F` against `t`, with the C-step step function; draw 0 at `t = 13` and the cell `t = 14` drawn as open lower bounds), `pfdY13_report.pdf`, `pfdY13_summary.json` (the three rows, the Amendment 14 check and the reading) and `scripts-archive.json`. The generator scripts are kept out of the repository, as in stage A, stage C and the t = 12 note, in a tarball (`scripts.tar.gz`) recorded in the benchmark record and stored in S3 by SHA-256. `scripts-archive.json` names it, with its SHA-256, size and S3 path.
- Rule text: `research/prime_fall_degree_20261027/PREREGISTRATION.md`, Amendment 7 B, D and E, Amendment 14.

7. Status of the scripts archive

`results/Y13/analysis/scripts.tar.gz` holds the scripts and their output (SHA-256 `73157c8bb102bbd9...`, 4845 bytes; `scripts-archive.json` has the full hash and the S3 path). It is an entry of the benchmark record above and is stored in S3 by SHA-256 (`tools/benchrec.py add-raw` and `upload`, run by the main session on 2026-11-10; the object was listed afterwards).